

Vinoth M

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Experience

Associate Software Engineer | Aptean

Sep 2024 - Present

Coimbatore, India

- Defined **HLD/LLD** solution designs, data flow specs, and **XML/WSDL/SOAP** & REST integration standards for 15+ carrier APIs (incl. Amazon Logistics, FedEx) on a logistics platform serving **50K+ daily requests** with **99.9% uptime**, collaborating with leads and architects.
- Architected a production **RAG system** using **Azure OpenAI** and **pgvector** semantic vector search, serving **8K queries/day** at **p99 latency of 450ms**, reducing engineer lookup time by **53%**.
- Engineered **multi-agent AI workflows** (Azure OpenAI) with **Chain-of-Thought** and **Few-Shot** prompting to parse specs, generate mapping configs, and trigger validation, cutting carrier onboarding effort by **70%** (2 weeks → 3 days).
- Designed a metadata-driven carrier mapping engine applying **SOLID** principles and **Factory + Strategy** patterns, enabling zero-code carrier onboarding and eliminating **40 hrs/month** of manual configuration.
- Implemented **OAuth 2.0** client credentials flow with **JWT** auth and **RBAC** across 14+ integrations; added **Redis** caching for carrier code lookups, cutting DB hits by **38%** and reducing mapping-workflow **p99 latency from 900ms to 350ms**.
- Developed **Java** backend services with **JDK 17+** (records, virtual threads, pattern matching) and **multithreading** for concurrent request handling; optimized React frontend via code-splitting, improving Time-to-Interactive by **40%** for **5K daily active users**.
- Maintained **GitHub Actions** CI/CD pipelines and deployed **Node.js** services on **Azure App Services** (Key Vault, Blob Storage); used the **AWS SDK** for **CloudWatch**/Elasticsearch monitoring, resolving 23+ production incidents and cutting **avg. resolution time by 30%**.
- Drove code reviews and resolved **SonarQube** static-analysis issues across a 10+ member team; coordinated Agile delivery with distributed teams spanning India, China, and the US, delivered sprint demos to stakeholders.

Projects

LinkedIn Content Agent | Node.js, Gemini API, Multi-Agent AI, React, OAuth 2.0

- Built a full-stack AI platform with a 3-agent pipeline (Planner, Writer, Reviewer) using Chain-of-Thought prompting to turn a topic into a publish-ready LinkedIn post; integrated **OAuth 2.0** authorization code flow with token lifecycle management and a React UI featuring live agent-reasoning traces and approval gates.

JobNest - Microservices Job Portal | React.js, Redux, Spring Boot, MySQL, Redis, JWT

- Engineered a job portal with **Spring Boot microservices** (auth, jobs, applications) secured with JWT and RBAC; designed a normalized MySQL schema with indexed queries reducing search latency by **35%**; built a React + Redux frontend with role-based views, lazy loading, and client-side caching.

DocReady - Document Analysis & Migration Readiness Tool | Python, Gemini AI, React, AWS SQS

- Built a full-stack tool analyzing .docx/.pdf documents for knowledge-base migration readiness; used **AWS SQS** to asynchronously queue analysis jobs for concurrent processing, with an LLM-powered verdict engine producing structured readiness scores and actionable improvement suggestions.

Education

Sri Krishna College of Engineering and Technology

Dec 2021 - May 2025

B.Tech in Information Technology CGPA: 8.0 / 10.00

Coimbatore, Tamil Nadu

- Activities:** Organized hackathons and codeathons, led department technical events, and mentored junior students in technical community initiatives.

Technical Skills

Languages: Java 17+, JavaScript, TypeScript, Python, SQL, C#

Backend / Frontend: Spring Boot, RESTful APIs, SOAP/WSDL, Microservices, Maven, Node.js, Express.js, React.js, Redux, .NET Core

AI / LLM: Azure OpenAI, RAG Systems, Multi-Agent AI, Prompt Engineering (CoT, Few-Shot), pgvector, Vector Search

Databases & Cloud: PostgreSQL, MySQL, SQL Server, Redis, AWS SDK, API Gateway, Lambda, ECS, EC2, RDS, S3, SQS, CloudWatch, Azure (App Services, Key Vault, Blob Storage), Docker, GitHub Actions, Elasticsearch

Messaging / Streaming: Apache Kafka, AWS Kinesis

Concepts & Tools: System Design (HLD/LLD), SOLID, Design Patterns, Multithreading, SonarQube, OAuth 2.0, OIDC, JUnit 5, JWT, Agile, Postman, Git

Achievements and Certifications

- Oracle Database @ AWS Certified Architect Professional (2025) | AWS Academy Cloud Foundations | Red Hat OpenShift Applications
- ReactJS & Redux - Udemy | Java SE 8 - Infosys | Python - Coursera
- Smart India Hackathon 2023 - qualified internal round (150+ teams) | LeetCode - 400+ problems solved